

Personal Programming Project Report

Title: Expense Tracker

1. **(5pts) Honor Code and LLM Usage for this Report.**

As a Hokie, I will conduct myself with honor and integrity at all times. I will not lie, cheat, or steal, nor will I accept the actions of those who do.

During the preparation of this assignment, Jai Sundaram used [name of the tool] in the Personal Programming Report to support the development and help explain it better. After using this tool, I/we reviewed and edited the content as needed to ensure its accuracy and take full responsibility for the content in relation to grading.

2. **(15pts) Learning Objectives:**

List the learning objectives from your proposal. In your own words explain whether you met those objectives and how (50-100 words each objective). Also describe if you learned something different than expected or anything additional.

Skill Development: Gain experience with version control systems like Git.

I believe that working on this personal programming project gave me significant exposure to version control systems like Git. I gained exposure to aspects of Git such as creating a repository, creating a Readme file, and organizing the repo by creating appropriate folders for the files in my project. Working on this project illustrated how it is crucial to integrate a version control system into the overall development workflow.

Skill Development: Develop proficiency in using Python for data analysis.

This assignment, in which I created a personal programming project, an expense tracker, helped me develop proficiency in using Python for data analysis by giving me exposure to working with real-world data and performing operations on it. I used Python to collect, store, and process the data. I performed numerous calculations, such as calculating totals and categorizing data into meaningful sections. Furthermore, I worked with the CSV module to read and write structured data, which is crucial in data analysis.

3. **(15pts) Timeline:**

Outline how you spent time on your project. Break down the time into specific tasks or milestones. Here is an adjustable schedule to get you started. Actual Details should be 50-100 words each and should compare or reflect on differences from your proposal.

Time	Task	Expected Details from Proposal	Actual Details
Hour 1-2	Research and gather resources	I will review basic Python concepts such as how to write a for loop, while loop, conditionals, user input, and displaying data. I will also see how to write and read data from a file.	I reviewed Python basics but focused more on how they apply to this project rather than just general syntax. I practiced loops and conditionals, and worked with input() to handle user interaction. I also experimented with the CSV module, specifically learning how to read and write structured data.
Hour 3-4	Design the project structure and plan	I will decide how the overall project will be built. I will decide what categories to include in my app, such as food, travel, entertainment, utilities, rent etc. I will also plan the functions that are needed, such as adding expenses, removing expenses, and modifying them, etc.	I finalized the structure of the program by deciding on how to structure the user experience. I chose to store expenses as dictionaries inside a list so they could be easily accessed and modified. I defined the main functions needed, including adding, viewing, deleting expenses, and generating summaries. I also decided to store all data in a CSV file to make sure the data is handled ideally.
Hour 5-6	Start coding the basic functionalities	I will begin coding and start off by creating a way for the user to enter their expenses, perform calculations, and make sure that all data is displayed properly.	I implemented the core functionality of the application. I created the functionality for users to add expenses. I added validation functions to ensure that the input data is proper. I also implemented viewing and deleting expenses, along with saving and loading data from a CSV file. I also ensured the menu system worked correctly

Time	Task	Expected Details from Proposal	Actual Details
			and that data was being stored properly.
Hour 7-8	Test and debug the initial version	I will test all possible use cases for the inputs. I will ensure that the calculations are accurate for all use cases. I will make sure to debug all errors.	I tested the program by trying different inputs, including invalid values and edge cases. I fixed issues related to incorrect inputs and made sure the program handled errors without crashing. I also debugged the overall file handling to ensure that expenses were correctly saved and loaded when the program restarted. I also verified that all options worked correctly.
Hour 9-10	Refine and add advanced features	I will refine the app and add more advanced features, such as categorizing expenses and totals for every category. I will also add functionality to store data to a file so that the user's data is not lost when the program quits.	I improved the program by adding features such as category summaries and monthly summaries to analyze spending patterns. I also created a budget checking feature that compares the total spending to a user-defined budget and provides feedback. I also added an export feature that generates a text report of all expenses. Finally, I improved the formatting of the output to make the program more readable and user-friendly.
Additional			

4. (55pts) Final Product Description:

Include your proposed MVP, Target, and Reach versions.

- i. Minimum Viable Product (MVP): The MVP is a CLI (Command Line Interface) based program that allows users to add, view, and delete [expenses](#). It will also calculate the spending

- ii. Target Product: This will be a more advanced version that involves budgeting, summaries, reports, and this version will also allow you to export.
- iii. Reach Version: The reach version would be an app that has a GUI, visualizations, and possibly more advanced versions to store the data.
- iv. (20pts) Description of final product, including target audience, user story, problem statement, key features, technical details, and technologies used. (100 – 150 words). This final product is a foundational expense tracker that helps users stay on top of their finances. The primary intended target audience is students and any other individuals who need additional help to manage their expenses. This application allows users to add and manage logs of their expenses, which will help them get a better understanding and improve their spending habits over time. A potential user story would be John, the freshman university student. Since he is living alone for the first time, he will need to manage his own finances, which may be initially hard to do. As a result, by using this app, he is able to keep a better track of it and improve his habits over time. Despite containing advanced functionality, the overall technical background of this program is straightforward. This program is built using Python and uses its functionality to conduct all tasks.
- v. (20pts) Provide a YouTube link to your video demonstration (1–2 minutes, narrated). **Important Note:** Do not upload your video file directly. Instead, upload your video to YouTube and include the video link clearly here in your report. The level of difficulty and detail of the project should be reasonable for 10 hours of work with LLM support. The project should not be something an LLM can solve without significant effort by the developer. (Be sure to have someone else test that your link is working.):
https://youtu.be/GkfTpev_cg0
- vi. (15pts) Any input files, coding files, and test files should be uploaded. Provide a list here of file names and purposes, or any links to live sites or artifacts. Remember code should also be commented. A README file should be created and uploaded so that we have the option to follow your instructions to run your project.
[tracker.py](#) - The overall program
Expenses.csv - A mock dataset
Expense_report.txt - Contains the exported data
Testcases.txt - Provides an overview of the test cases

Project Repository & Code Submission Details:

Project Repository on git.cs.vt.edu (other options are [code.vt](#):

<https://code.vt.edu/> or **(Git-hub only if you are part of virtual global collaboration)**): Your repository should be well-organized, documented, and

easy to navigate. At a minimum, include the following structure:

- **code/** – All source code files for your project (organized by component or module if applicable).
- **data/** – Any input files, datasets, or configuration files used by your program.

- **tests/** – Test scripts or files demonstrating how your code was verified.
- **docs/** – Supporting materials such as screenshots, reports, or documentation.
- **report/** – This final report document.
- **README.md** – A detailed file describing:
 - Project overview and purpose
 - Video link of your project
 - Installation and setup instructions
 - How to run the program and reproduce results
 - Technologies or libraries used
 - Author(s) and contribution summary

Required:

- Maintain a logical directory structure, do not store all files at the root level.
- Include comments in your code to explain logic and design decisions.
- Keep your repository **private** until grades are released, then you may make it public.

Share access with the following personnel (Add them as collaborators):

GTA Name	Section(s)	Professor
Mona Moghadampanah	13392 & 13394	S. Cao
Katelyn Crumpacker	13393 & 13395	M. Ellis

5. (10pts) Consultation and Use of LLMs:

Each student must create a unique project but is allowed to consult with other people and use Large Language Models (LLMs). Describe how you incorporated these resources into your project:

- **Consultation Description:**
Describe how you ended up seeking advice or feedback from peers, mentors, or online communities.
- I asked my friends if this was a problem that was intriguing and worth tackling. Furthermore, I showed them the program once completed and received feedback. Furthermore, I used online resources to overcome any developmental difficulties and debug.
- **Use of LLMs:**
Explain how you ended up utilizing LLMs to assist with coding, debugging, learning technologies and concepts, or generating ideas.

I used an LLM to help me structure my program overall. I used it to also create some mock data and asked it for the best way to approach the testing aspect. Furthermore, I integrated its use of it during overall development, having it explain complicated concepts and assist with debugging.